KIRANKUMAR ADAM

PERSONAL INFORMATION

Born in India, 28 September 1984

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OBJECTIVE

To be good Sensor Data and Information Fusion Engineer.

WORK EXPERIENCE

Jan 2023–Present Lead Perception & Autonomy, Asteria Aerospace Limited — Bangalore, India

Asteria

- Managing a team size of 5 Perception Engineers.
- Working and mentoring team on Visual SLAM, Sensor Fusion and 3D Reconstruction for verticals in Argo Tech, Telecom and Surveillance.
- Developing Sensor Fusion and Planning Algorithms for inhouse developed Drone and Quadcopters.

Mar 2022–Oct 2022 Lead Data Scientist, Aimlytics Technologies Private Limited — Hyderabad, India

Aimlytics

- Managed a team size of 4 Computer Vision engineers.
- Worked on Face Recognition, Anti-spoofing techniques and Behavioural Analysis of a person for internal clients.
- Worked on Speaker Recognition using SpeechBrain Toolkit for internal clients.
- Software packages: OpenCV, Keras, Pytorch, SpeechBrain.

Aug 2021–Mar 2022 Machine Learning Engineer, Turing Enterprise Inc. — (Remote) Palo Alto, California

PI Machine Learning and Computer Vision Consultant for Pi Network

- Worked on OCR for KYC mandate in multilingual settings using EasyOCR.
- Worked on Face Recognition and Anti-spoofing techniques for KYC mandate.
- Software packages: Python 3.7, OpenCV 4.3, EasyOCR.

April 2021–July 2021 Lead Machine Learning Engineer, Newspace Research and Technology Private Limited. — Bangalore

NRT Machine Learning Engineer for UAVs (Behavior Planning)

- Worked on Behavior Planning in UAVs using Behaviour Trees.
- Software packages: BehaviorTree.CPP 4.0 with C++ 17 standards.

June 2020– Dec 2020 **Tech Lead, L&T Technology Services Ltd. — Bangalore**

L&T Worked at client Huawei India Pvt. Ltd for Computer Vision and Machine Learning applications.

- Project Holosens
  - Implemented Image Augmentation and Synthetic Image Generation on the given image dataset.
  - Implemented Metric analysis for object classification in image dataset.
  - Software packages: Python 3.7, OpenCV 4.2, scikit-learn 0.23, keras 2.4.0

Dec 2019– June 2020 **Consultant, Deutsche Welthungerhilfe e.V. — Remote**

WHH

- Child Growth Monitor
  - Sensor Data Fusion for RGB and Point Cloud Data capture on child.
  - Classify child as Malnourished based on Z-scores calculated on weight, height, age and gender of a child .
  - Software packages: Python 3.7, scikit-learn, python-pcl

June 2018– Dec 2019 **Senior Engineer, Great Wall India Research and Development Private Limited — Bangalore**

GWM Engineer for Perception, Mapping and Localization

- OpenDrive HDMap Generation for Mapping and Localization team based on Point Cloud Data as per Baidu Apollo specifications.
- Multiple Object tracking and Track Management using Extended Kalman Filter and Global Nearest Neighbor methods.
- Software packages: OpenCV 4.0, Qt 5.14, Baidu Apollo 2.5 with C++ 11 standards.

Jan 2015– May 2018 **Research Consultant, Mindborn Software Solutions — Solapur**

Mindborn Research Consultant for clients in the area of Mathematical Modeling, Computer Vision and Estimation Theory:

- Object Detection using Saliency.
  - Cognitive Vision approach to detect salient object in an environment.
  - This work is used in two applications for our clients: State space tracking using EKF and Region Based Image Retrieval.
  - Software package: OpenCV 3.4.0, Boost libraries with C++11 standards.
  - Code: https://github.com/kiranadam/vocus2
  - Duration: April 2015 to November 2015.
- Anomaly detection in Diamond and its classification.
  - Working on Clarity aspect as per GIA specifications for anomalies as feathers, patches etc., for feature extraction.
  - Software package: OpenCV 4.0, Boost 1.65 libraries with C++11 standards.
  - Duration: December 2015 to Present.
- Extended Object Tracking using State Space approach.
  - Designing algorithms for Extended Object Tracking using Gaussian Processes.
  - Collaborative work with former colleagues at University of Bonn.
  - Software package: Matlab 7.6

April 2014–Nov 2014 Research Intern for Sensor Data and Information Fusion, VALEO Schalter und Sensoren GmbH — Bietigheim-Bissingen

Valeo Worked as a Research Intern for Sensor Fusion in ADAS system using LED sensor and mono vision camera.

- Worked on EuroNCAP for automated parking.
- Worked on Mono Vision and LED object level fusion for Multi-target tracking and Fusion applications using EKF and GNN algorithms.
- Software package: Matlab 7.6

June 2012–Feb 2013 Research Assistant for Sensor Data Fusion, Bonn University — Bonn

Bonn Worked as a Research Assistant and Teaching Assistant for Sensor Data and Information Fusion Department.

- Mathematical Modeling and Algorithm development for Extended Object Tracking using Random Matrices.
- Worked on Optical flow and Mean Shift based algorithms for drones in tracking an object of interest.
- Software package: OpenCV, Eigen with C++ 11 standards.

Sep 2010–Jan 2011 Consultant for United Nations, UNFCCC — Bonn

Bonn Worked as Software Consultant and Data Assimilation Engineer for United Nations Framework Convention on Climate Change for the project TTClear.

June 2010–Sep 2010 Research Assistant for University Hospital, Bonn University — Bonn

Bonn Worked as a Research Assistant for Machine Learning Applications for Dept. Applied Mathematical Physiology Laboratory under Prof. Dr. Sven Zenker

EDUCATION

July 2023–Ongoing International Institute of Information Technology, Hyderabad

PhD Robotics Research Center · Electronics and Communication Engineering Department
Area of Working: *Sensor Fusion and GPS Denied Navigation Systems*
Advisors: Prof. Harikumar Kandath · harikumar.k@iiit.ac.in

Jan 2015 University of Bonn, Bonn

Masters of Science Computer Science · Institute of Computer Science
Thesis: *Sensor Fusion using LED and Vision Sensor*
Grade: 3.00/4.00
Advisors: Prof. Juergen Gall · gall@iai.uni-bonn.de

Mar 2008 Walchand Institute of Technology, India

Bachelor of Engineering Computer Science and Engineering · Solapur University
Grade: 66%
Advisors: Prof. Raj Kulkarni · raj_joy@yahoo.com

SKILLS

Subjects	Sensor Data Fusion · Computer Vision · Machine Learning · Robotics · ADAS · Estimation Theory
Programming	C/C++ · Java · MATLAB · Python · Scala
Libraries	OpenCV · Breeze · Keras · Scikit · CUDA · Aramadillo · Eigen
Languages	English, Hindi, Marathi, Telugu, German
Interests	Reading · Cooking · Puzzle Solving

November 8, 2024